AWS | PROJECT

VPC Peering Project Report

Connect two VPCs. Prove private connectivity.

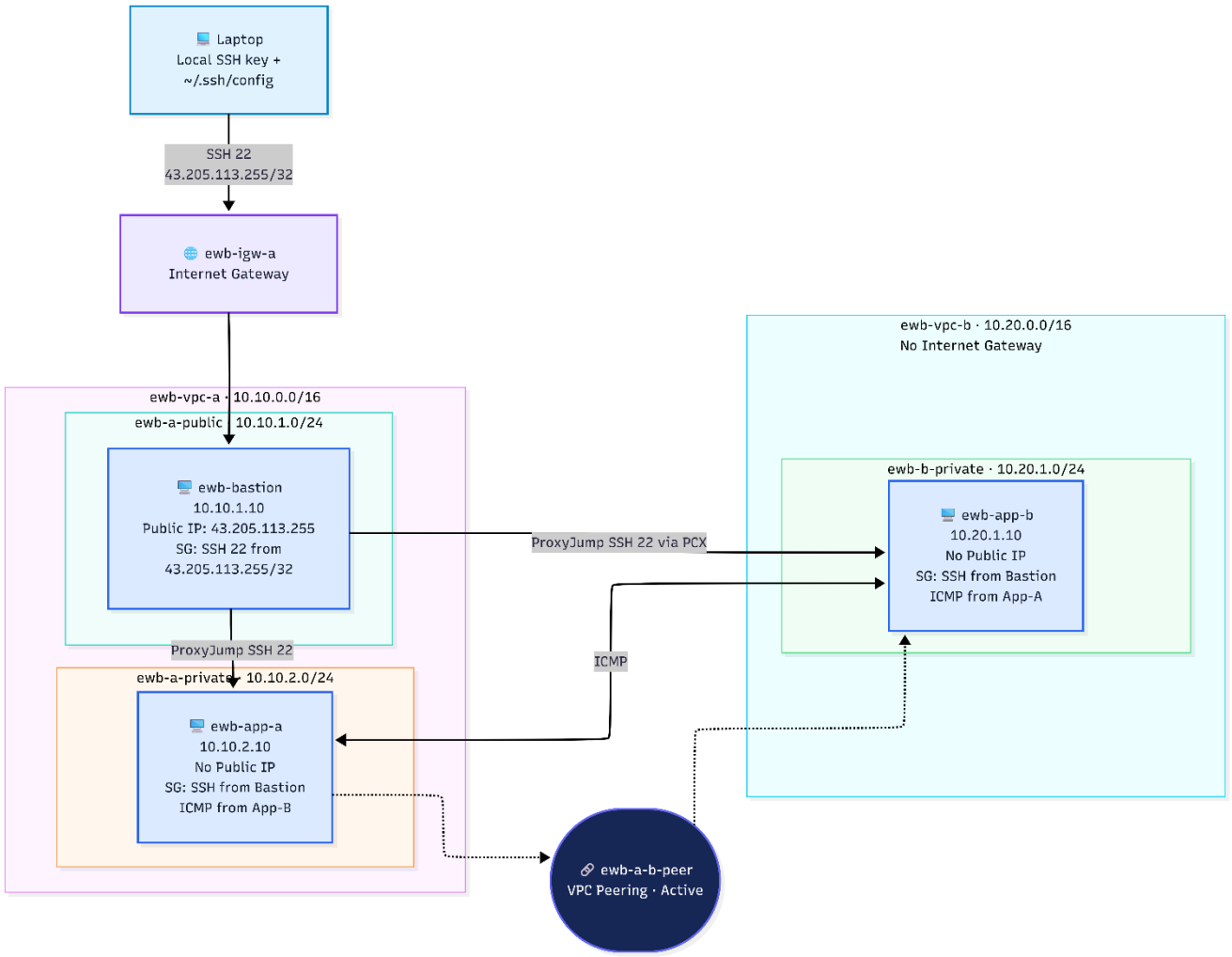
Student name	Kavya R
Batch	1
Project title	AWS VPC Peering: private connectivity between two VPCs
AWS Region / AZ	Mumbai / ap-south-1
Completion date	18/10/2026
GitHub repository URL	https://github.com/Kavyarajeevs/ewb-vpc-peering
Instance type / AMI	Amazon Linux 2023

1. Summary

5-8 lines: goal, what you built, result of the tests. (Marks area: report clarity, 10)

I built two isolated VPCs in ap-south-1: VPC A (10.10.0.0/16) with a public and a private subnet, and VPC B (10.20.0.0/16) with one private subnet. A single bastion host in VPC A's public subnet is the only internet-facing instance. App-A and App-B have no public IPs. Before peering, App-A could not ping App-B (100% packet loss). After creating and accepting the peering connection 'ewb-a-b-peer' and adding routes on both sides, App-A and App-B pinged each other with 0% loss over private IPs, and I reached App-B through the bastion using SSH ProxyJump. A controlled test showed that removing App-B's ICMP rule stopped the ping and restoring it fixed it. I then deleted all resources and verified nothing remained.

2. Architecture and resource inventory



Inventory (actual IDs)

Resource	Name tag	ID	CIDR / IP
VPC A	ewb-vpc-a	vpc-038316686e3fd79c8	10.10.0.0/16
Public subnet	ewb-a-public	subnet-0f9d07f78b20ce771	10.10.1.0/24
Private subnet A	ewb-a-private	subnet-052099c51ff85409f	10.10.2.0/24
Public Route Table A	ewb-rt-a-public	rtb-0d9a542b84f6b4d12	
Private Route Table A	ewb-rt-a-private	rtb-0dac6f215be11025c	
VPC B	ewb-vpc-b	vpc-0f62f07e9bbe3db94	10.20.0.0/16
Private subnet B	ewb-b-private	subnet-0c1329f736d171c25	10.20.1.0/24
Private Route Table B	ewb-rt-b-private	rtb-07b99b444b1e68f3e	
Internet gateway	ewb-igw-a	igw-0c08cbcb3b451d6f2	attached to VPC A
Peering	ewb-a-b-peer	pcx-046305deb7a5add5e	A to B
Bastion	ewb-bastion	i-01a47a3dd2ba1e204	10.10.1.10 / public: 43.205.113.255___
App-A	ewb-app-a	i-0c7203ccf92eaa15f	10.10.2.10 / none
App-B	ewb-app-b	i-04b421322faf3cfcf	10.20.1.10 / none

Why the CIDRs do not overlap (2-3 lines):

10.10.0.0/16 and 10.20.0.0/16 are separate address ranges, so every IP belongs to exactly one VPC. Peering needs this so a route like 10.20.0.0/16 has one unambiguous destination, and AWS refuses to create a peering connection between overlapping VPC CIDRs. Each /24 subnet sits inside its own VPC’s range.

3. Network build: subnets, internet gateway, route tables

CHECKPOINT 01

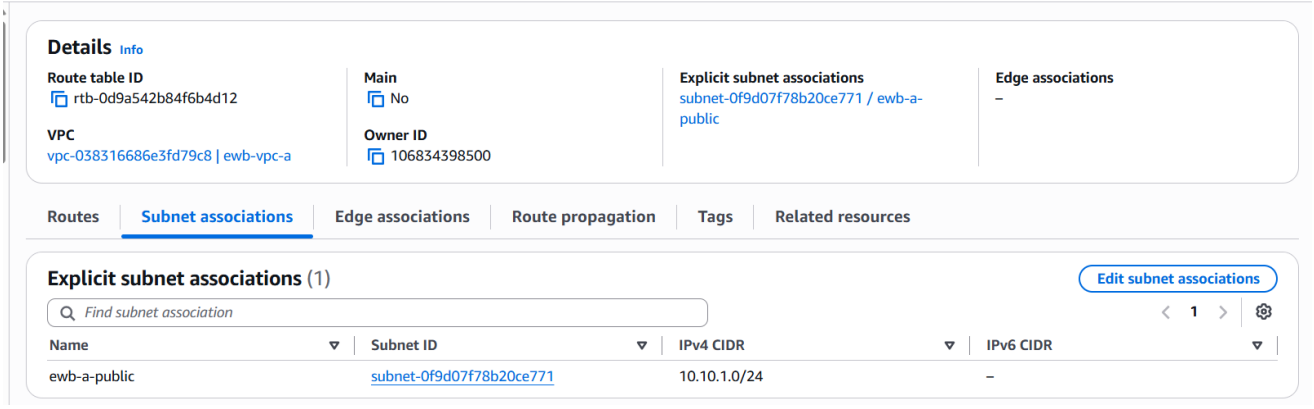


Figure 1: Subnet Association of route table A public

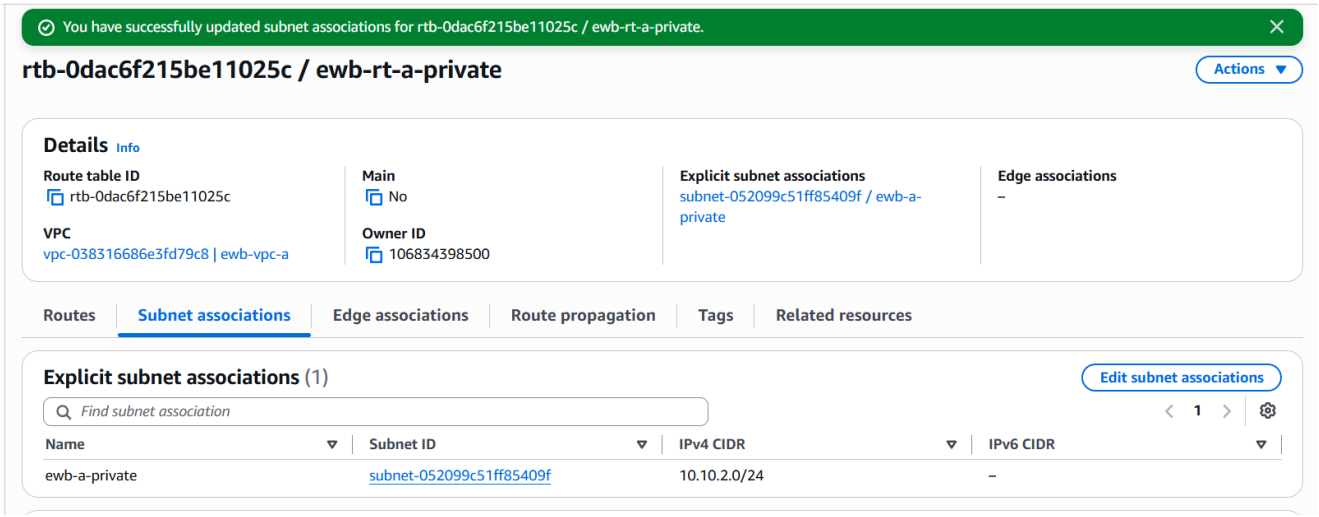


Figure 2: Subnet Association of route table A private

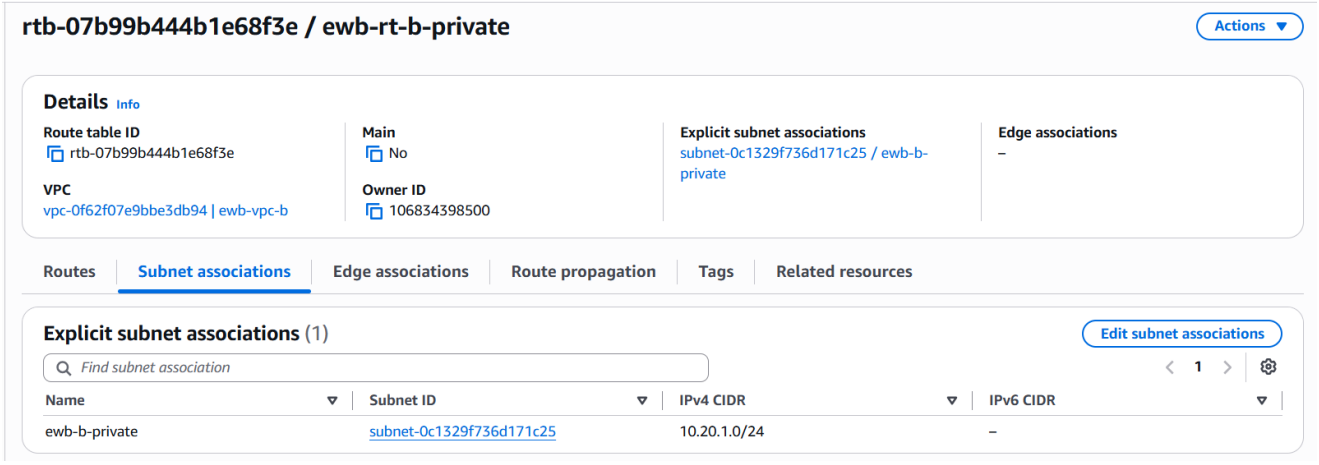


Figure 3: Subnet Association of Route Table B private

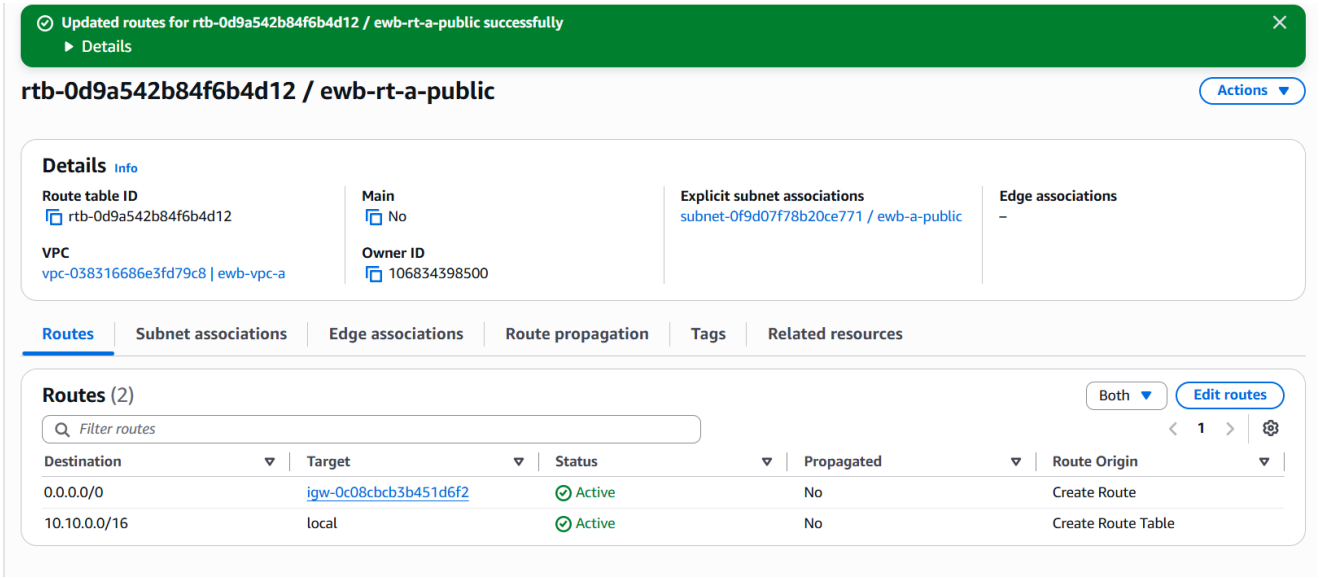


Figure 4: public default route

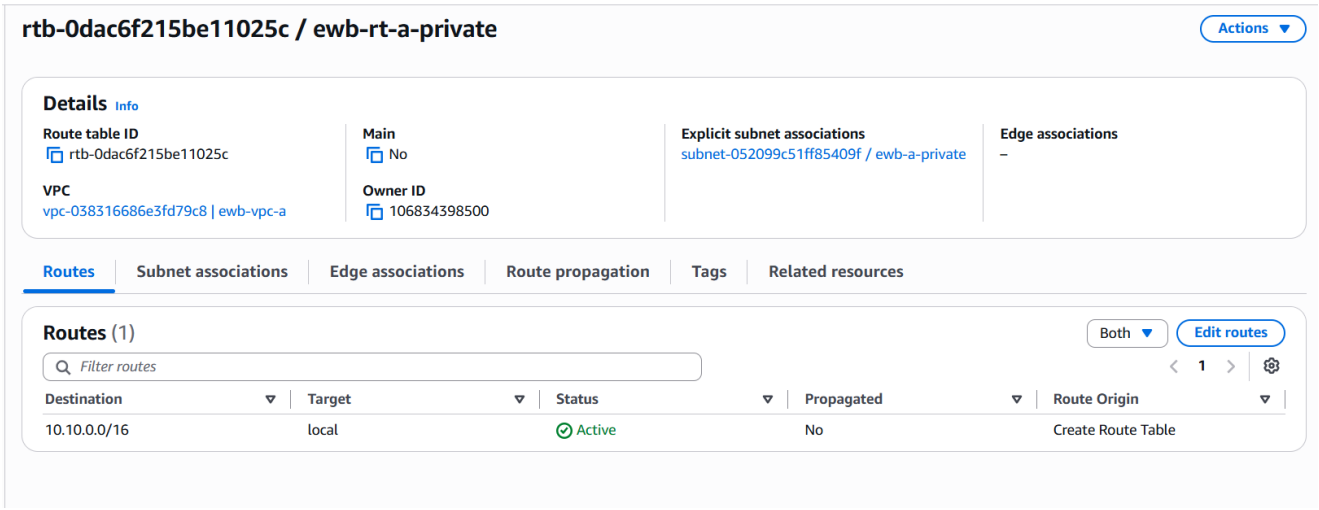


Figure 5: ewb-rt-a-private shows only the local route (before peering)

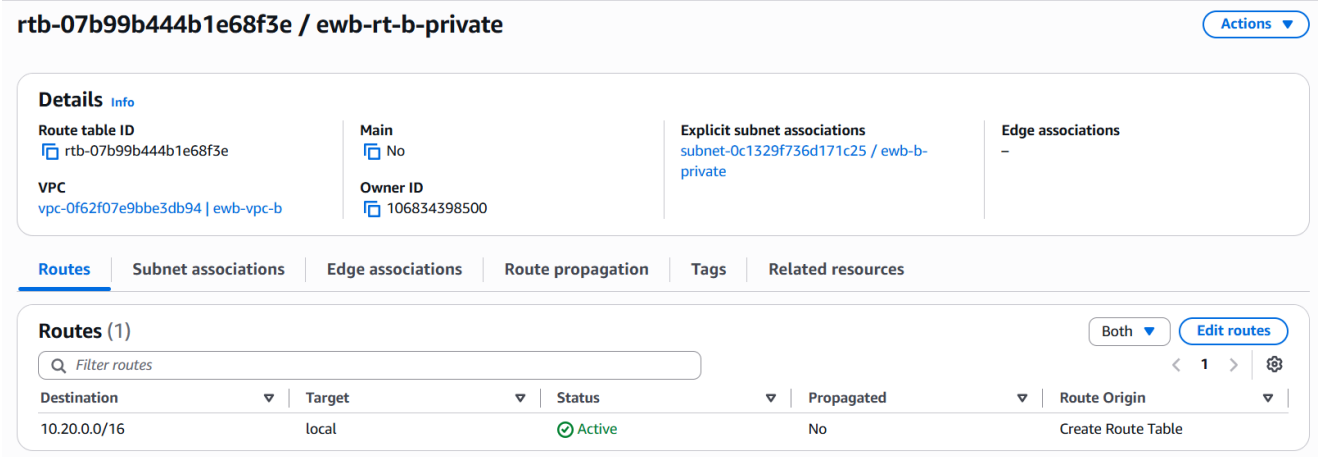


Figure 6: Route Table B Private

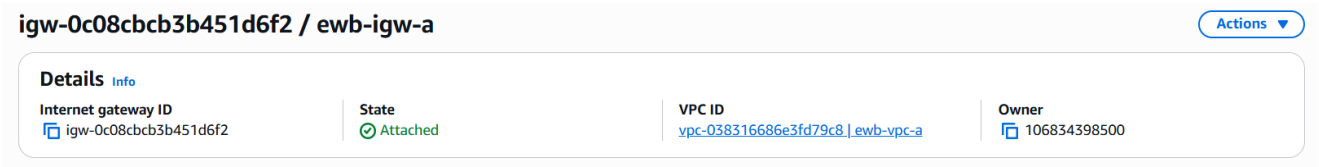


Figure 7:Internet Gateway Attached to VPC A

Why is a subnet 'public'? Why does the private subnet have no IGW route?

A subnet is public because its route table has a 0.0.0.0/0 route to an internet gateway; the instance also needs a public IPv4 address to be reached directly. ewb-a-private has only the local route and no IGW route, so its instances cannot send or receive traffic directly from the internet.

4. Security groups

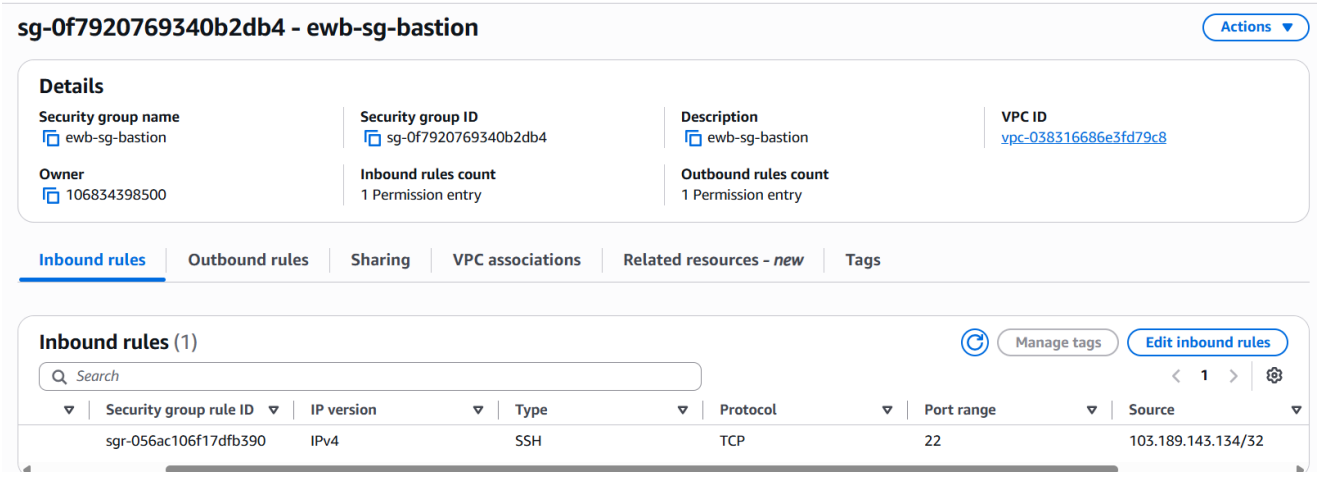


Figure 8: Inbound rules: ewb-sg-bastion

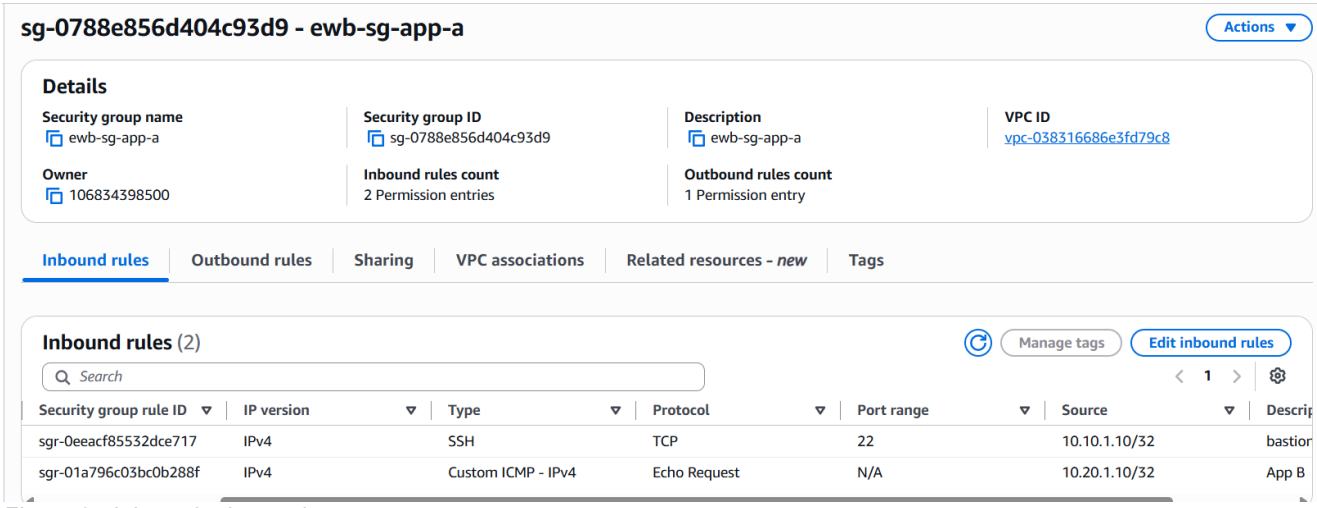


Figure 9:: Inbound rules: ewb-sg-app-a

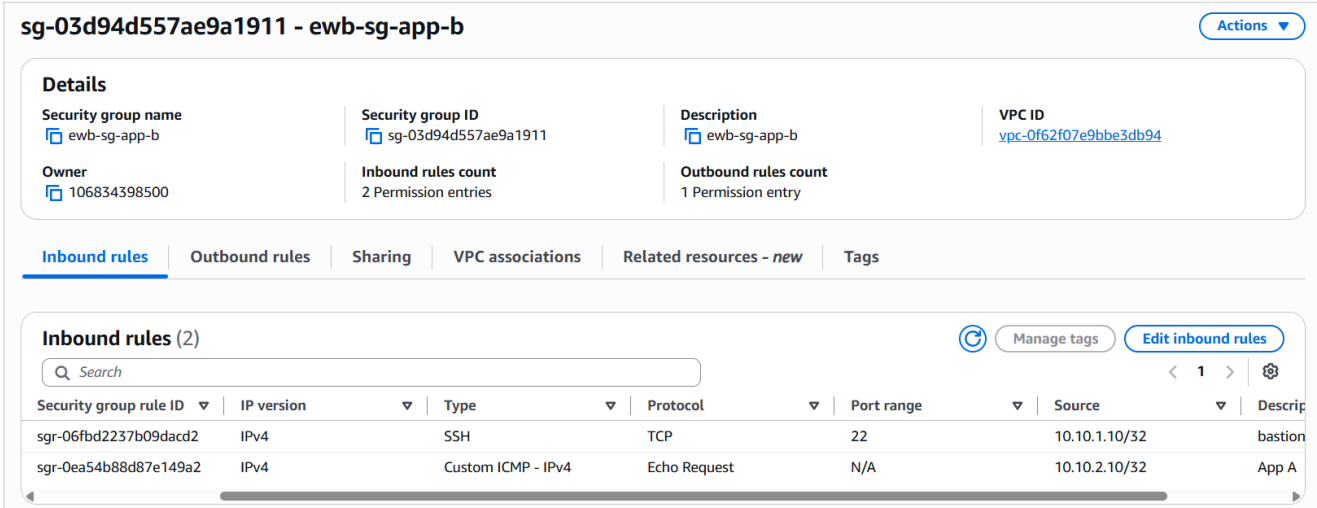


Figure 10: T: Inbound rules: ewb-sg-app-b

5. Instances and SSH access

CHECKPOINT 02

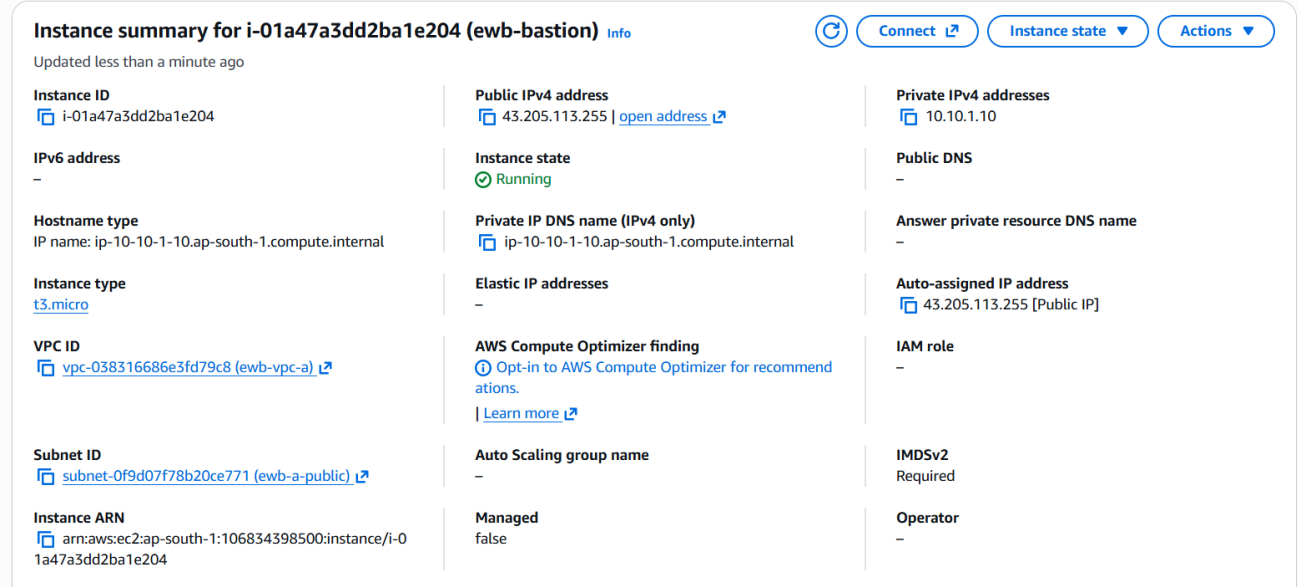


Figure 11:ewb-bastion

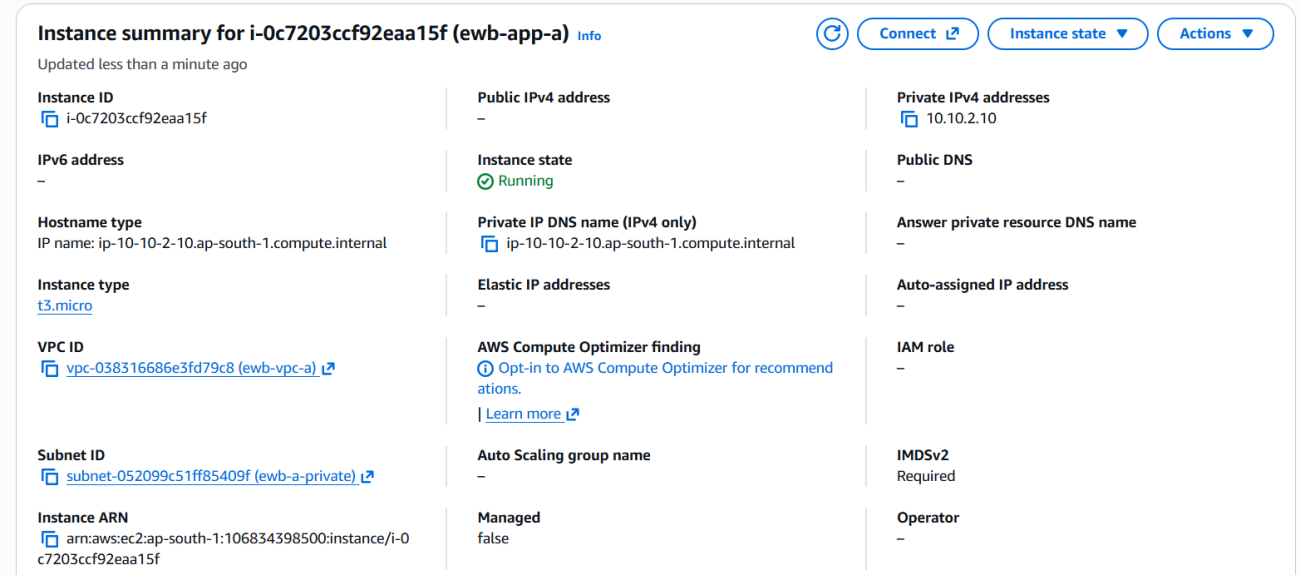


Figure 12:ewb-app-a Public IPv4 address field empty

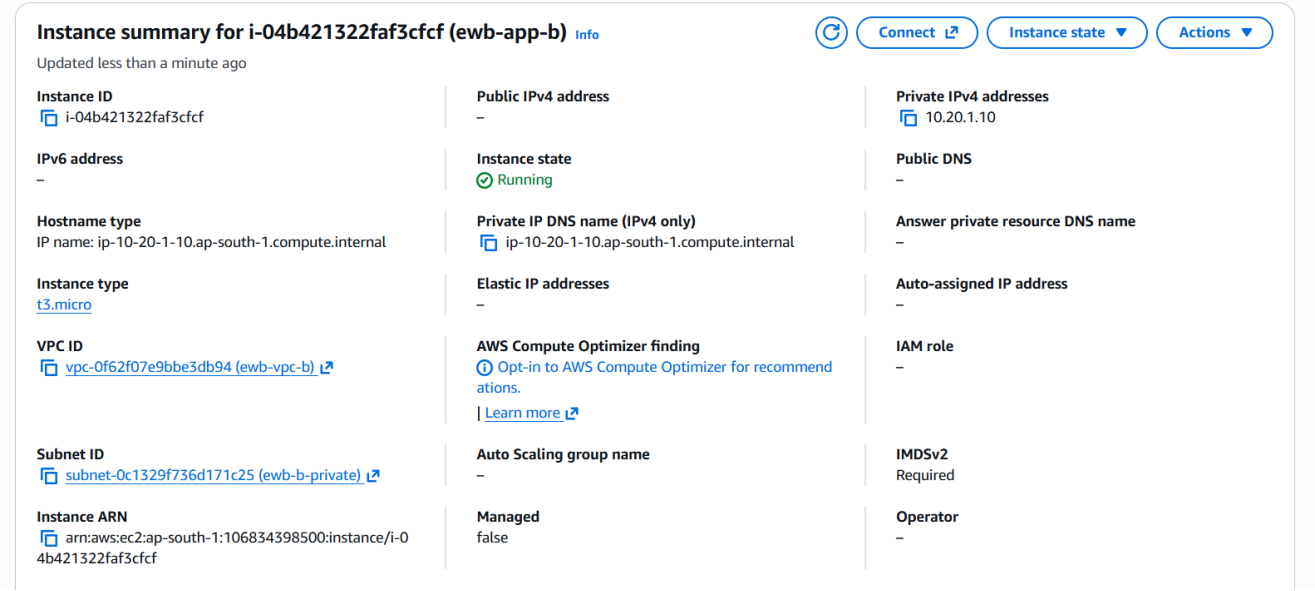


Figure 13: ewb-app-b Public IPv4 address field empty

~/.ssh/config

```
laptop: cat ~/.ssh/config
```

```
 Host ewb-bastion
    HostName 43.205.113.255
    User ec2-user
    IdentityFile ~/.ssh/ewb-lab.pem
    IdentitiesOnly yes

Host ewb-app-a
    HostName 10.10.2.10
    User ec2-user
    IdentityFile ~/.ssh/ewb-lab.pem
    IdentitiesOnly yes
    ProxyJump ewb-bastion

Host ewb-app-b
    HostName 10.20.1.10
    User ec2-user
    IdentityFile ~/.ssh/ewb-lab.pem
    IdentitiesOnly yes
    ProxyJump ewb-bastion
```

```
PS C:\Users\hp> ssh ewb-bastion
The authenticity of host '43.205.113.255 (43.205.113.255)' can't be established.
ED25519 key fingerprint is SHA256:+RixdLpOFbIpAwLyPt0R3jtc8/30rM6qFeZHcB84EQ0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '43.205.113.255' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-10-1-10 ~]$ hostname -I
10.10.1.10
```

Figure 14: ssh ewb-bastion

```
PS C:\Users\hp> ssh ewb-app-a
The authenticity of host '10.10.2.10 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:Mp8JTXDn1PgyZCJaXm4VfL7qyT5VUc8ZYQMCrUaKohw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.10.2.10' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-10-2-10 ~]$ hostname
ip-10-10-2-10.ap-south-1.compute.internal
[ec2-user@ip-10-10-2-10 ~]$ hostname -I
10.10.2.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3117ms

[ec2-user@ip-10-10-2-10 ~]$ exit
logout
Connection to 10.10.2.10 closed.
```

Figure 15: Baseline. Before peering and peer routes, it received no replies

6. Peering connection and routes

CHECKPOINT 03



Figure 16: ewb-a-b-peer, Active, both VPC IDs (requester A, accepter B)

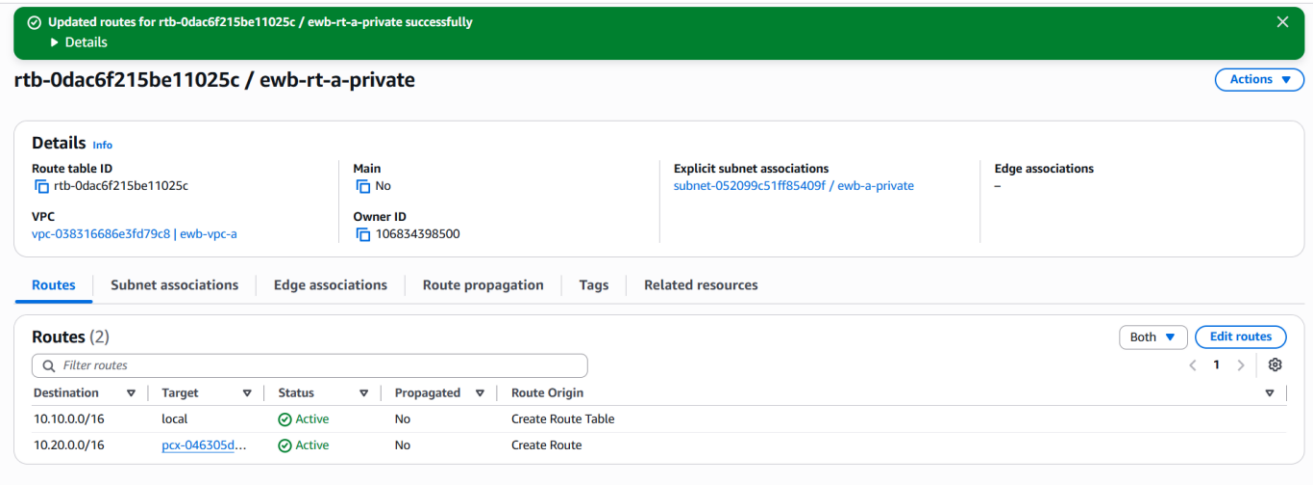


Figure 17: ewb-rt-a-private / associated to ewb-a-private

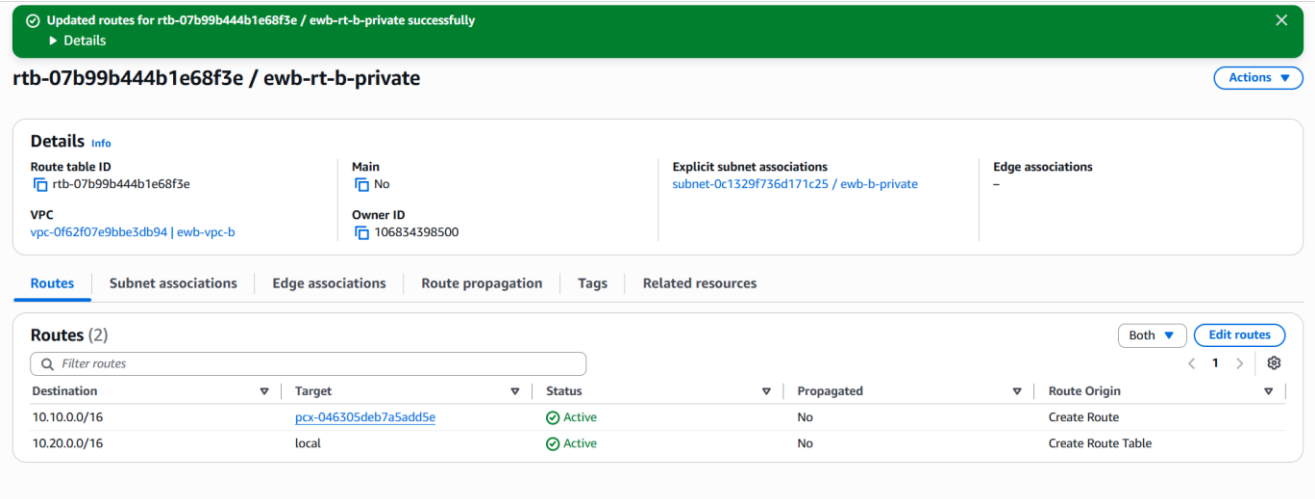


Figure 18: ewb-rt-b-private / associated to ewb-b-private

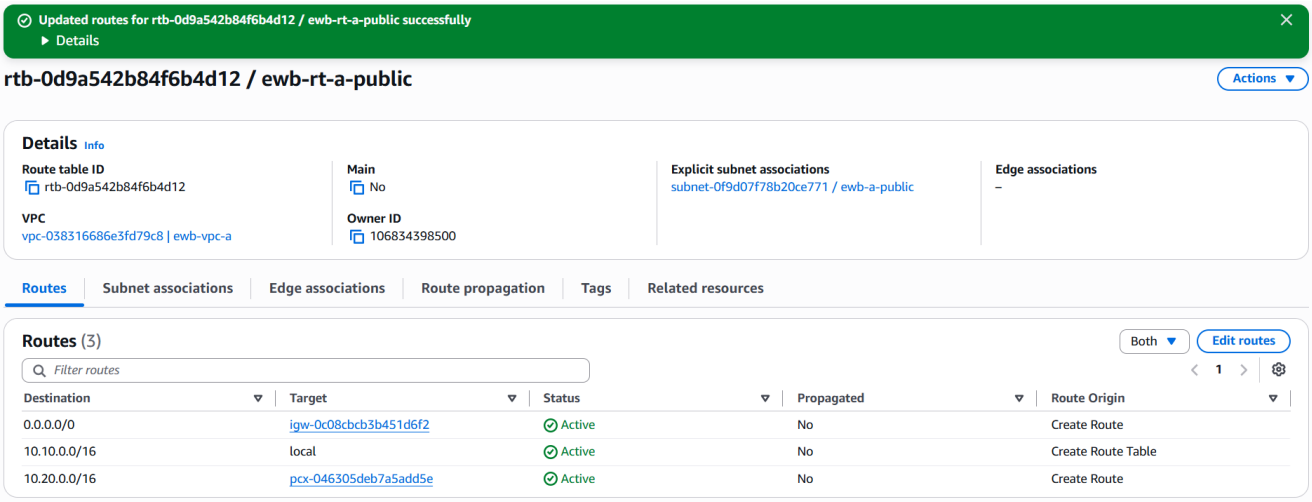


Figure 19: ewb-rt-a-public / associated to ewb-a-public

7. Test evidence

Test	Host / command	Expected	Actual (PASS/FAIL)
01 Baseline	App-A: ping -c 4 10.20.1.10 (before peering)	No replies	PASS
02 Forward	App-A: ping -c 4 10.20.1.10	4 replies, 0% loss	PASS
03 Reverse	App-B: ping -c 4 10.10.2.10	4 replies, 0% loss	PASS
04 Private SSH	Laptop: ssh ewb-app-b	Shell via bastion	PASS
05 Isolation	EC2 details App-A / App-B	No public IPv4	PASS
06 Controlled failure	Remove App-B ICMP rule; ping; restore; ping	Fail then recover	PASS

Test 01 | App-A | baseline (hostname, hostname -I, ping)

```
PS C:\Users\hp> ssh ewb-app-a
The authenticity of host '10.10.2.10 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:Mp8JTXDn1PgyZCJaXm4VfL7qyT5VUc8ZYQMCrUaKohw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.10.2.10' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-10-2-10 ~]$ hostname
ip-10-10-2-10.ap-south-1.compute.internal
[ec2-user@ip-10-10-2-10 ~]$ hostname -I
10.10.2.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3117ms

[ec2-user@ip-10-10-2-10 ~]$ exit
logout
Connection to 10.10.2.10 closed.
```

Test 02 | App-A | forward path

```
PS C:\Users\hp> ssh ewb-app-a
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Sun Oct 18 10:58:50 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.456 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.165 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.172 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.180 ms

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3109ms
rtt min/avg/max/mdev = 0.165/0.243/0.456/0.122 ms
[ec2-user@ip-10-10-2-10 ~]$ exit
logout
Connection to 10.10.2.10 closed.
```

Test 03 | App-B | reverse path

```
PS C:\Users\hp> ssh ewb-app-b
The authenticity of host '10.20.1.10 (<no hostip for proxy command>)' can't be establish
ed.
ED25519 key fingerprint is SHA256:bOEu8OgWUm2jZSvuFWomtBob7Kyr4yHOUasrRG4cS/s.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.20.1.10' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-20-1-10 ~]$ ping -c 4 10.10.2.10
PING 10.10.2.10 (10.10.2.10) 56(84) bytes of data.
64 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=0.182 ms
64 bytes from 10.10.2.10: icmp_seq=2 ttl=127 time=0.183 ms
64 bytes from 10.10.2.10: icmp_seq=3 ttl=127 time=0.138 ms
64 bytes from 10.10.2.10: icmp_seq=4 ttl=127 time=0.178 ms

--- 10.10.2.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3108ms
rtt min/avg/max/mdev = 0.138/0.170/0.183/0.018 ms
[ec2-user@ip-10-20-1-10 ~]$ exit
logout
Connection to 10.20.1.10 closed.
```

Test 04 | App-B | SSH

```
PS C:\Users\hp> ssh ewb-app-b
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Sun Oct 18 11:29:19 2026 from 10.10.1.10
[ec2-user@ip-10-20-1-10 ~]$ hostname
ip-10-20-1-10.ap-south-1.compute.internal
[ec2-user@ip-10-20-1-10 ~]$ hostname -I
10.20.1.10
```

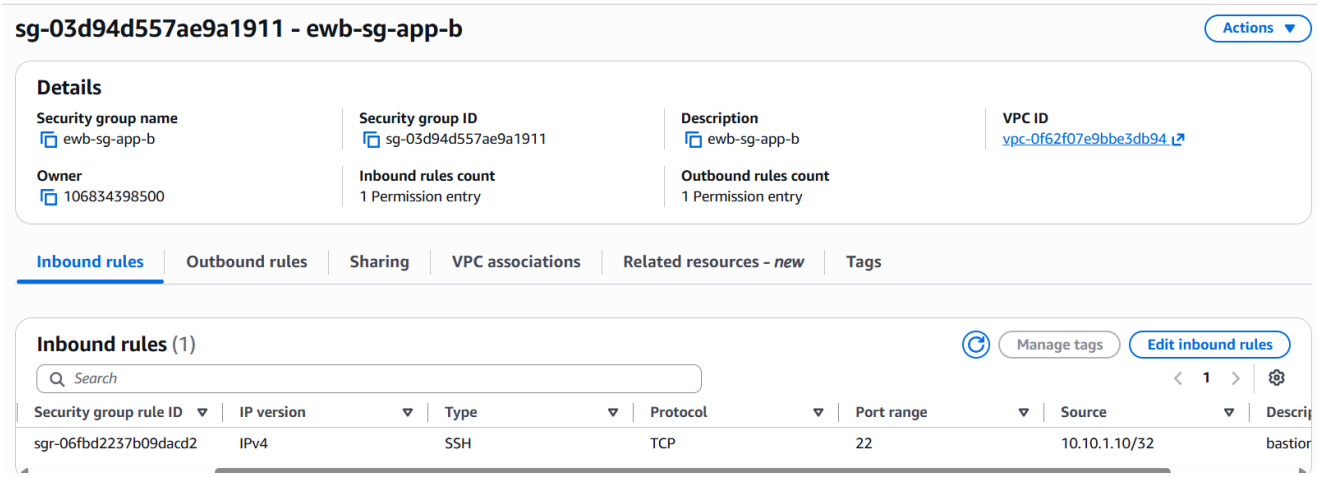


Figure 20: App-B ICMP rule REMOVED

```
PS C:\Users\hp> ssh ewb-app-a
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Sun Oct 18 11:22:18 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$ hostname -I
10.10.2.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3139ms
```

Figure 21: ping failed

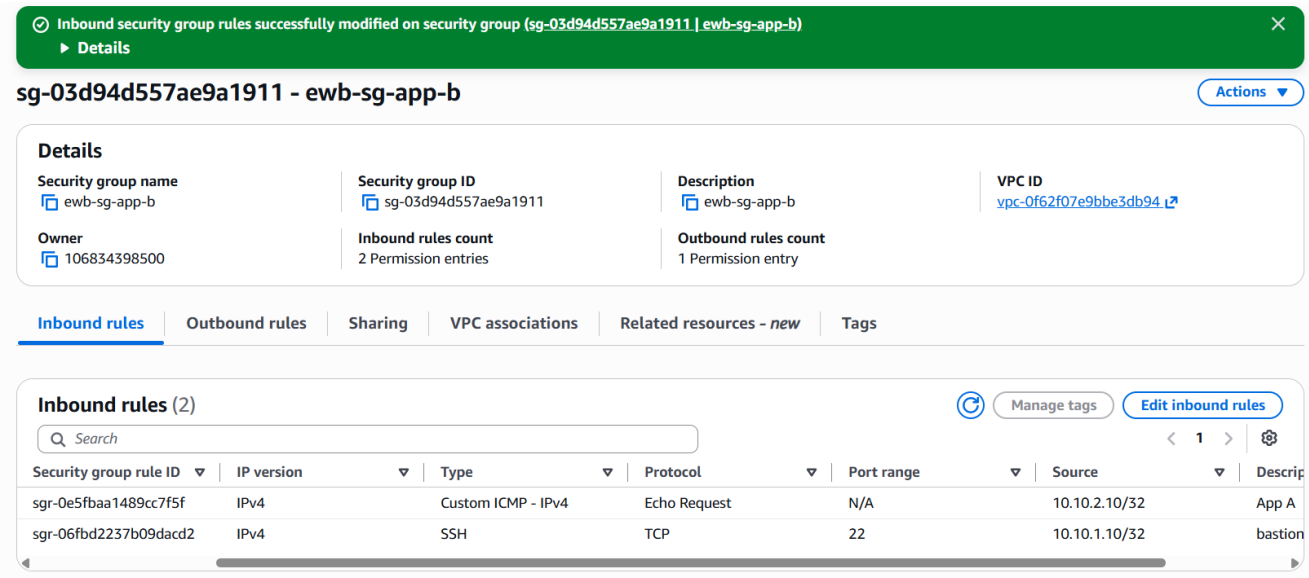


Figure 22: Rule restored

```
PS C:\Users\hp> ssh ewb-app-a
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Sun Oct 18 11:45:03 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.169 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.163 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.220 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.173 ms

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3120ms
rtt min/avg/max/mdev = 0.163/0.181/0.220/0.022 ms
[ec2-user@ip-10-10-2-10 ~]$ exit
logout
Connection to 10.10.2.10 closed.
```

Figure 23: Ping succeeds

Explain the controlled failure: which layer failed and why were routes/peering not the cause?

I removed only App-B's inbound ICMP echo request rule from 10.10.2.10/32. The ping from App-A then showed 100% loss, and after I restored the identical rule it returned 0% loss. Peering, route tables and the SSH rule were unchanged throughout, so the security group on the destination was the layer that blocked the traffic, not routing. This shows that Active peering with correct routes is not enough: the destination security group must also allow the traffic.

8. Forward path and return path explanation

Forward path: App-A (10.10.2.10) to App-B (10.20.1.10)

The echo request leaves App-A, and the default outbound rule allows it. ewb-rt-a-private matches 10.20.0.0/16 and sends it to the peering connection. It arrives in VPC B's subnet, and ewb-sg-app-b allows it because the source is 10.10.2.10/32 and the type is echo request. The default network ACLs allow it too.

Return path: App-B to App-A

Security groups are stateful, so App-B's reply is allowed automatically. The reply is still routed, though: ewb-rt-b-private matches 10.10.0.0/16 and sends it to the peering connection, which delivers it back to App-A. Without that route the reply has nowhere to go.

Bastion path to App-B (ProxyJump)

With ProxyJump, the bastion opens a new SSH connection to App-B, so App-B sees the bastion's private IP 10.10.1.10 as the source, not my laptop. ewb-rt-a-public therefore needs 10.20.0.0/16 to the peering connection, and it is chosen over the 0.0.0.0/0 internet route because it is more specific.

Viva questions

Question	My answer
Why is the return route needed?	Packets must be routable in both directions. Without a route from VPC B back to 10.10.0.0/16, replies are dropped even though the request arrived.
Why is App-B's SSH source the bastion private IP?	The bastion starts the second connection, so App-B sees 10.10.1.10. Using that /32 limits SSH to the bastion only.
Why does Active peering not guarantee ping?	Peering only creates the link. Routes, security groups and network ACLs still decide whether traffic can use it.
Can VPC B use VPC A's IGW over peering?	No. Peering is one-to-one and does not allow routing through the other VPC's internet gateway, so VPC B stays without internet access.

9. Troubleshooting log

No unexpected failures. The baseline and the controlled ICMP test were the two intentional failures.

10. Cleanup evidence

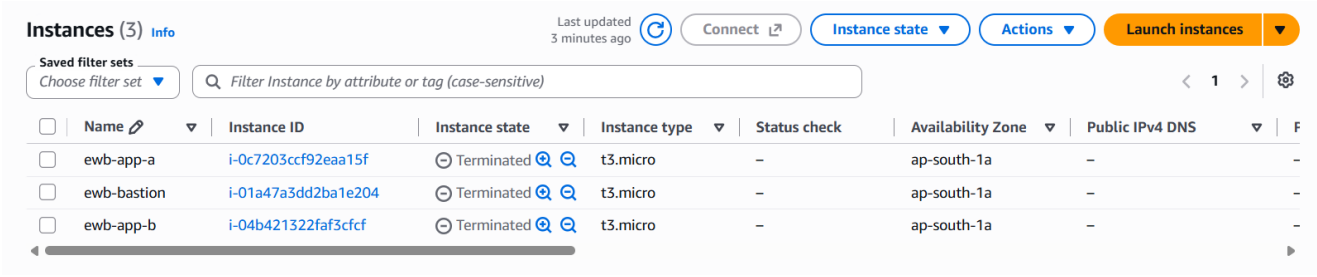


Figure 24: Terminated 3 instances

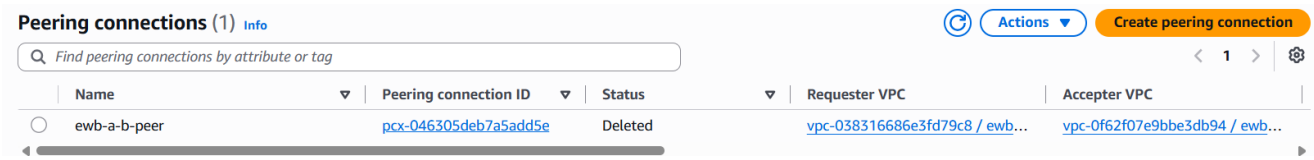


Figure 25: deleted peering connection

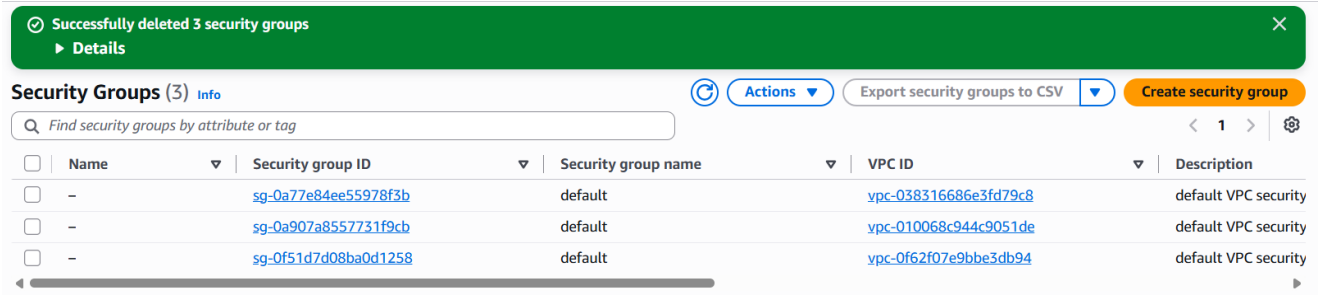


Figure 26: deleted three security groups

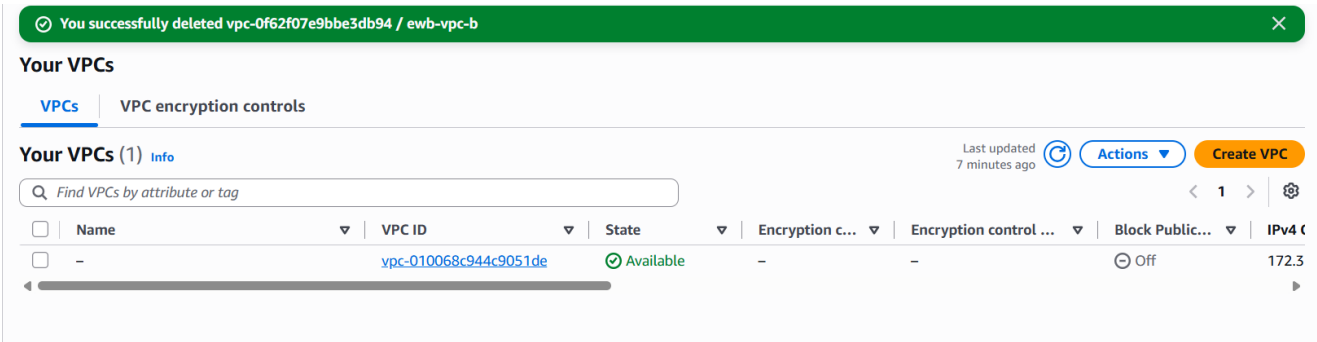


Figure 27: Deleted 2 VPCs